INCH-POUND

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15 February 1989

COMMERCIAL ITEM DESCRIPTION

TEST SET, FUEL SYSTEM

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID), describes a Bridge, Capacitance/Resistance for a Test Set (BC/RFGTS). This CID is meant as a minimum requirement for the TSFS in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type II, class 2, style B and color R in accordance with MIL-T-28800. This CID includes test specifications which makes provisions for the bid sample testing in accordance with MIL-T-28800.

APPLICABLE DOCUMENTS

Specifications and standards The following specifications and standards form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800 Test Equipment for use with Electrical and Electronic Equipment, General Specification for.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Other publications The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE STD 587-1980

AMSC N/A FSC 6625

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(Application for copies of IEEE publications should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

Order of precedence In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the type II safety requirements of MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage.

2. Parts, materials, and processes. Except for marking, the Bridge, Capacitance/Resistance for a fuel gage test set shall comply with the type II parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a Restricted material The equipment shall comply with the restricted material requirements of MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type II design and construction requirements of MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Type of construction. Construction of the BC/RFGTS shall be of a monolithic or a separate component design.

b. Modular design restrictions. The modular design concept for the TSFS is acceptable with the following restrictions:

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(1) The BC/RFGTS transit case shall consist of two separately molded pieces, a housing for the FGSTS and a hinged, detachable cover.

(2) Each module shall contain a power circuit protection fuse.

(3) Each module plug-in card shall be keyed and mated with the back plane receptacle so that modules may not be randomly placed into the mainframe.

(4) Module shall not be removable by access through the front or rear of the mainframe without the unaided hand.

(5) The TSFS shall come with one(1) blank space cover and one(1) extender board.

(6) A package part number shall be assigned to the Bridge, Capacitance/Resistance for a fuel gage test set as a whole system. This system part number shall be marked on the mainframe identification plate and all plug-in(s).

(7) All module interface shall be done in the back of the mainframe.

c. Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be Type BNC female.

d. Internal cooling. The equipment shall be capable of operating continuously for twenty four (24) hours at 55°C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the BC/RFGTS when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts ac (±10%) at line frequencies of 50 Hz to 400 Hz.

a. Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

b. Power consumption. Maximum power consumption of all components combined shall not exceed 40 watts.

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c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

d. Surge and transient voltage protection. The BC/RFGTS shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per Type II solid state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to IEEE Std 587-1980. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the BC/RFGTS may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal Type II steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the BC/RFGTS shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the BC/RFGTS shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. The size and weight of individual components shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment dimensions. The nominal dimensions per component for the TSFS should be approximately 18 inches long x 15 inches wide x 13 inches deep.

b. Equipment weight. The weight including all accessories (excluding manuals) per component, shall not exceed 25 lbs.

6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the style A enclosure requirements of MIL-T-28800.

a. Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

b. Tilt bail handle. A tilt-bail handle shall be provided in lieu of the hinged metal handles in accordance with MIL-T-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

c. Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

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7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

(1) Panel marking. The front panel marking shall comply with the type II parts, materials and processes panel marking requirements of MIL-T-28800; however, marking identification processes other than the Type II requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

b. Items to be marked (if applicable).

(1) Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure.

(2) The type designation of the AN nomenclature of the BC/RFGTS shall be stenciled on the exterior or main frame on at least three adjacent surfaces, on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the Class 2 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -51°C to +71°C.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of -20°C to 55°C.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 2 equipment as specified by MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 2 equipment as specified by MIL-T-28800.

d. Vibration requirements. The equipment shall meet the class 2 vibration requirements as specified by MIL-T-28800.

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e. Mechanical shock requirements.

(1) Shock, functional. The equipment shall meet the class 2 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

f. Electromagnetic interference (EMI). The equipment shall comply with MIL-T-28800.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type II, class 2 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 and the upper test shall be MTBF 4,000.

b. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) BURN-IN. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40°C for a minimum of 100 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be conducted at selected times during the burn-in test.

(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 2 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

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a. Maintainability. The equipment shall meet the type II maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type II interchangeability requirements specified by MIL-T-28800.

12. Digital interface. The equipment shall meet the digital interface requirements per MIL-T-28800.

13. Performance characteristics. Unless otherwise specified, all vernier controls shall be in the calibrated position and all performance requirements shall apply with or without the probes, after a 30 minute warm-up period in a laboratory ambient operating temperature.

a. General specifications. The test set shall provide a means to determine the serviceability of aircraft employing an AC capacitance-type fuel gaging systems. It shall consist of a single integral unit incorporating a transit case, direct current (dc) electrical resistance measuring circuitry, capacitance measuring circuitry, capacitance substitution circuitry (simulates tank unit and compensator capacitance), calibration circuitry with built-in standards of capacitance, switches, wiring, cable assemblies, and accessories. The test set shall operate from an external power source and/or a dry cell battery source. The structure and feature relating to calibration shall provide for lowest cost, simplicity, accuracy and reasonable convenience in use.

b. Test set functions. The test set shall contain four basic types of circuitry. Each circuit shall perform a separate function as described.

(1) Capacitance measurement. The test shall have AC capacitance measurement to determine the serviceability of aircraft capacitance type fuel quantity gaging systems. The test set shall automatically display the measured capacitance (i.e. no necessary computation of the tank unit and the compensator. The capacitance measurement system shall have a range from 0 to 8000 pf, the zero measurement may be relaxed slightly but not more than 5 pf. The measured capacitance value shall have an accuracy of plus or minus 0.10% of the maximum value of each capacitance range. The values established up scale and down scale shall not deviate from that accuracy by more than 0.05%. The resolution shall be at least 0.025% of the range selected.

(2) Simulated tank unit and compensator capacitance. The BC/RFGTS shall be able to simulate tank units capacitance in the range from 0 to 8000 pf and simulate compensator capacitance in the range from 0 to 390 pf. The capacitance simulation shall have an accuracy of 0.10% of reading.

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(3) Insulation and tank resistance measurement. The resistance measurement module shall have a range from 0.1 to 100K megohm. The resistance measured shall not deviate from the true resistance by more than 15%. The resolution shall be at least 0.01% of the range. The DC voltage applied across the circuit shall not exceed 50 volts and the short circuit current, when the output receptacles are shorted together shall not exceed 0.010 amperes.

(4) Calibration. The BC/RFGTS shall have built-in capacitance which shall be accurate to 0.15% for capacitance of 50 pf or more and 0.1% for capacitance less than 50 pf. The capacitance and resistance standards shall be traceable to the National Bureau of Standards. Calibration adjustments shall be provided to permit precise settings at zero and maximum ends of the scale. All circuits shall have inherent stability built in to preclude the need for frequent recalibration. Sufficient adjustment range shall be provided to allow for calibration. The time required for calibration shall not exceed 45 minutes for a knowledgeable technician (not including the 0.5 hour warm-up period).

c. Power requirements. The TSFS shall meet all voltage and frequency requirements as set forth in the MIL-T-28800. A dry cell battery shall provide operating power when power is not available (flight line use). The battery shall be rechargeable and shall be able to supply power for a minimum of 8 hours of continuous operation. If non-rechargeable batteries are used, they shall be required to provide a minimum of 40 hours continuous operation.

d. Cables and connector types. Universal cabling and adapters shall be used wherever practical. Relocation of cables shall not be required when changing modes of operation (i.e. from capacitance to resistance).

(1) A pair of heavy-duty leads with spade or banana type connectors on one end and spring clips with protective sleeves on the other shall be furnished for measurement of resistance. The cable shall be 15 feet in length plus or minus 6 inches. The BC/RFGTS shall have binding posts for facilitating the measurement of resistance externally.

(2) All aircraft tank units and indicator connectors shall be BNC type.

(3) A power cable shall be used for charging of the battery and/or bench top use.

e. Indicators. All measurements shall be displayed on a 5-digit (minimum) lighted indicator with floating decimal point. The indicator shall be visible in bright sunlight and meet all requirements of class 2 of the MIL-T-28800.

f. Documentation. The manufacturer shall include the calibration procedures and specifications as well as the operation and maintenance manuals.

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14. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Packaging, packing, and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping container shall be marked as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Known acceptable products: The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Part Number	NSN
GULL	361-015-001	6625-01-035-0257

Custodian
Air Force - 99

Reviewer:
GSA-FSS

User:
Navy - SH

Preparing activity:
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